MAIN

SCSP Hackathon 2026 - Round 1 - Hacker 1-Pager

Here, you will find everything you need to know about this event. If you have questions beyond this document, please reach out to hack@scsp.ai or reach out to our lead organizer, James at jdr@scsp.ai.

Phase 1: Saturday April 25 – Sunday April 26, 2026 · SF / DC / Boston
Phase 2: Saturday May 9, 2026 · AI+ Expo, Walter E. Washington Convention Center

Find and Register a Team:

You may register in teams of up to 5 hackers. You may also compete solo. If you're still looking for team members, or looking to add a member to your team, please find them in-person or in our discord server!

After the start of the hacking, you have until 2pm on Saturday, April 25th, local time to finalize your team. By that time, we expect to have received an email to hack@scsp.ai from your team announcing your final team and track. Please CC all of your team members on this email to confirm their participation on your team.

Subject: SCSP Hackathon <Team Name> FINAL
Body:
Our team name is [team name].
Our track is [insert track].
Our members are:

- FirstName1 LastName1
- FirstName2 LastName2
- FirstName3 LastName3
- …

**Note that you must register your team before 2 PM on Saturday and after this deadline you may not change your team.*

Contact

General inquiries: hack@scsp.ai
Live Zoom Office Hours: email James Ryseff to request a consultation
Technical / hack inquiries: email James Ryseff (jdr@scsp.ai)

Prize Breakdown:

The top team of each track in each location will move onto Phase 2 on May 9, 2026. At Phase 2, each team will present before a panel of independent judges, who will determine final rankings for each track.

Prizes will be awarded per track as follows:
1st place: $10,000
2nd place: $5,000
3rd place: $2,500

Total Prizes: $4\ tracks\ \times\ (\$10,000 + \$5,000 + \$2,500) = \$70,000$

Note: Certain government officials and other disqualified persons (Schmidt-entity employees or contractors, substantial contributors, family members of those individuals, etc) will be ineligible to receive any prize money offered by SCSP. If you believe you may fall into one of these categories, please let the SCSP team know and we will gladly review your situation. SCSP reserves the right to determine prize eligibility in its sole discretion. Individuals deemed ineligible may participate but will not receive prize funds.

Judging Rubric:
Judging Criteria- Scoring Rubric (1-10 scale per category):

Novelty of Approach (25%) Does this solution bring a fresh perspective? Judges will evaluate how creative and original the team's approach is — whether they challenged conventional thinking, introduced a new method, or tackled the problem from an unexpected angle.

Technical Difficulty (25%) How technically ambitious is the build? Judges will assess the complexity of the technology used, the sophistication of the implementation, and whether the team pushed beyond off-the-shelf solutions to build something genuinely challenging.

Potential National Impact (25%) Could this solution scale? Judges will consider how meaningfully this product could affect communities, systems, or people across the country — prioritizing solutions that address widespread problems and have a credible path to broad reach.

Problem-Solution Fit (25%) Our goal is to have our technical-minded folks match up with subject-matter experts. We are asking, does the team truly understand who they're building for? Judges will evaluate how well the team has identified and articulated a real user need, and how directly and effectively their product addresses that problem.

Please reach out to hack@scsp.ai with any questions about the judging criteria.

All submissions will be evaluated exclusively by independent judges with no affiliation with SCSP. Judges for the first round of judging are listed on each track-specific page. Final round judges will be announced prior to that round. Judging will take place from 5:00 PM - 7:00 PM local time, with results to be announced via email shortly after. Prizes are awarded based on merit as determined by the judging rubric. All judging decisions are final and binding.

Entry Submission:

You must create your project in a public GitHub repo (which must remain public throughout the course of the project). Then, you must submit your GitHub link along with a README.md file to hack@scsp.ai at 5:00 PM your local time with the following email template:

Subject: SCSP Hackathon <Team Name><track>
Body:
Github link: https://github.com/example_user/example_repo/
Attached README.md file containing (required):

- Team names
- Track
- What you built
- What datasets/APIs you used
- How to run it.

Note that at least 1 person on your team must be able to attend the demo of your project in-person. Failure to meet presentation requirements may result in disqualification from judging or prize eligibility. **Your project presentation should be kept under 5 minutes of demo time.**

Should you qualify for Phase 2 at AI+ Expo, you may not edit your Round 1 submission leading up to your presentation in Round 2.

Tracks:

(Please see the track tabs on this document)

Schedule:

Saturday, April 25
10:00 AM - 11:00 AM: Registration, find a team
11:00 AM: Hackathon kickoff.
1:00 PM: Must vacate premises
2:00 PM: Team and Track final email due

Sunday, April 26
5:00 PM: Submissions due

Other:

- You may only register with one team.
- Your team may only compete in one track.
- SCSP reserves the right to modify, suspend, or cancel the hackathon in the event of unforeseen circumstances.
- SCSP reserves the right to disqualify participants for misconduct or rule violations.

FAQ:

Q: Is vibecoding allowed?
A: Yes. Use whatever tools make you faster: Cursor, Claude Code, Copilot, Codex, Windsurf, v0, anything. Judges care about what you built, not how you typed it.

Q: Is async / remote hacking allowed?
A: Yes. Hack from anywhere. You just need one teammate onsite for the final presentation.

Q: What are the requirements for team sizes?
A: Teams should consist of 1–5 members. If you are attending solo, you can form or join a team onsite during the registration period.

Q: Is there a digital space for real-time support and team-finding?
A: Yes, you can join the hackathon Discord for real-time Q&A, dataset help, and to find teammates. The invite link will also be provided at the registration desk. You can also mingle with your fellow hackers at the hackathon locations

SCSP Hackathon Discord Link

Q: How can I get more context or specific details on a track?
A: You can reach out to hack@scsp.ai, or contact James (jdr@scsp.ai) to clarify your questions over Email / Zoom.

Q: Can I switch tracks mid-hackathon?
A: Technically yes, but you'll lose hours. Pick well during the 11:00 AM track announcements. The final deadline to submit your official team is 2pm local time.

Q: Can I use my own datasets or proprietary data?
A: Yes, as long as you have the rights and your project can be made public without exposing sensitive information. Judges must be able to see the project run during the evaluation.

Q: Do I need a security clearance for the GenAI.mil or Wargaming tracks?
A: No. All organizer-provided data is unclassified and public. Please do not bring any classified or restricted materials to the event.

Q: Are there any restrictions related to ITAR, export controls, or sensitive technologies?
A: Yes. Participants should be mindful of applicable U.S. export control laws and regulations, including the International Traffic in Arms Regulations (ITAR) and Export Administration Regulations (EAR). Projects should use only unclassified, publicly available, or otherwise shareable data and materials, since submissions will be made public. Avoid including or relying on any controlled or restricted information. Teams are responsible for ensuring that their work does not involve export-controlled or sensitive dual-use technologies in a manner that would violate applicable laws or regulations. Participants must have a determined export classification for preexisting code incorporated into their project.

Q: Who owns the IP of my final project?
A: You will retain ownership of your code and IP created during the hackathon. SCSP receives a non-exclusive license to make the work product publicly available. SCSP will publish all submitted projects – including (if available) title, description, tech stack, slide links, Github repositories, and participant names – on our GitHub page in an event-specific repository.

Q: How will prizes be paid?
A: Prizes will be distributed evenly among all eligible members of the winning team. Each participant will receive an equal share of the total prize amount associated with their team's placement. Payments will be made directly to individual participants, subject to verification of eligibility and compliance with all applicable rules and requirements. Prizes will be paid within a reasonable period after SCSP has received all necessary documentation (such as a completed W-9 and banking information). Winning participants are responsible for any applicable taxes or reporting obligations related to prize payments.

Q: Who should I contact if I have questions during the hackathon?
A: You can email hack@scsp.ai, or for more technical questions, you can contact James at jdr@scsp.ai.

Q: Where can I find SCSP's Event Privacy Notice?
A: Right here! SCSP Event Privacy Notice

Good luck. Build something the country needs.

Autonomous Labs

Autonomous Laboratories

AI is transforming how science gets done: not just analyzing data, but running experiments. Build an agent that automates any part of the scientific process.

The idea: Science has always been a loop: guess, test, learn, repeat. Your job is to use AI to speed up or automate some part of that loop. This could mean automating tedious lab workflows, helping researchers design better experiments, analyzing results faster, or building a full autonomous hypothesis-to-conclusion pipeline. The scope is wide, if it removes friction from how science gets done, it counts.

A good starting point: Think about what a lab technician or researcher actually does all day: pipetting, logging results, cross-referencing literature, reformatting data, writing up protocols. Pick one of those tasks and automate it. If you're not sure what's painful in a real lab, ask one of our subject matter experts: they've spent time in research environments and can point you toward high-value targets.

Judges:

SF	Boston	DC
Tuan Ho	Paul Hayre	Tuan Ho
Sriharsha Makineni	Sriharsha Makineni	Jeremy Baum
Tanish Dhanjal	Tanish Dhanjal	

Example project directions:

- Autonomous experiment loop: An agent takes a hypothesis, queries or simulates relevant data, interprets the outcome, and proposes the next experimental variation.
- AI data stent for broken pipelines: A lightweight adaptive layer detects workflow or schema drift and either repairs the pipeline or produces a precise fix.
- Protocol automator: take a natural language description of an experiment and generate a step-by-step lab protocol (or Opentrons script)
- Literature synthesizer: ingest papers from a domain and surface contradictions, open questions, or next experiments worth running
- Drug repurposing agent: query PubChem for known compounds, use AlphaFold structures to assess binding potential, iterate toward candidates
- Abstraction layer: develop the software layer that allows AI agents to move data and control physical instruments across different sites and without manual intervention.
- Operating system: Just as cloud computing environments have an operating system to manage their storage and compute resources, cloud laboratories will need a sophisticated resource orchestrator capable of managing multi-tenancy in a physical

environment. An intelligence scheduler should balance the need to optimize overall throughput of the laboratory, ensure a fair distribution of laboratory time among multiple potential users of the lab, while also detecting and recovering from errors which may occur in intermediate steps of the experiment.

Datasets & APIs (all free, most no-auth):

- **Materials Project:** Best for autonomous materials discovery loops, candidate ranking, inverse design demos, and active-learning style next-step selection. API docs | REST API overview | Python client docs | GitHub
- **Open Reaction Database + ORDerly:** Best for chemistry-focused agent loops, reaction-condition optimization, and structured reaction prediction tasks. ORD docs | ORD site | ORD schema GitHub | ORDerly GitHub
- **protocols.io public protocols:** Best for natural-language-to-protocol systems, step parsing, ambiguity detection, and escalation logic. protocols.io | Developer portal | API docs
- **WorkflowHub + RO-Crate:** Best for pipeline monitoring, workflow discovery, compatibility checks, and provenance-aware orchestration. WorkflowHub | GA4GH TRS | RO-Crate overview | ro-crate-py
- **Frictionless Data:** Best for schema-drift detection, validation, and automated repair suggestions in data-stent projects. Validation guide | PyPI package
- **Optional extensions: OpenML and NCBI Datasets:** Useful for teams that want quick ML benchmarking or biology-focused public datasets. OpenML Python docs | OpenML Python GitHub | NCBI Datasets docs | NCBI Datasets GitHub

Repos and SDKs:

- mp-api: Official Materials Project Python client and development repo.
- ord-schema: Schema, validators, and tooling for Open Reaction Database datasets.
- ro-crate-py: Python library for creating and consuming RO-Crates.
- frictionless: Python package and CLI for schema and data validation.
- openml-python: Quick path to datasets, tasks, and experiments for ML benchmarking.
- NCBI Datasets: CLI and APIs for pulling public sequence, genome, and metadata packages.

Primary links: mp-api | ord-schema | ro-crate-py | frictionless | openml-python | NCBI Datasets

Electric Grid

Electric Grid Optimization

AI data centers are spiking electricity demand at a scale the grid wasn't designed for. Build agents that keep the lights on.

The idea: The US power grid is being pushed to its limits. Your challenge is to build AI agents that can forecast demand, coordinate supply, and keep the grid stable in real time. Think of it as air traffic control, but for electrons. You don't have to solve the whole grid — it is just as valid to find one lever worth pulling and pull it hard.

A good starting point: Pull live or historical load data from EIA or CAISO. Build a demand forecaster first (even a simple regression baseline counts). Then layer in a supply optimizer that shifts loads or dispatches generation. Stability monitoring can be as simple as flagging when frequency drifts out of bounds.

Judges:

SF	Boston	DC
Dustin Janatpour	Dustin Janatpour	Monty McGee
		Dr. Masoud Barati

Example project directions:

- Data center demand coordinator: model how large GPU clusters should throttle workloads based on real-time grid signals
- Renewable integration planner: use solar/wind forecasts from NREL to schedule storage dispatch and minimize curtailment
- Grid anomaly detector: train on historical load patterns to flag instability before it cascades
- Infrastructure planning: identify key bottlenecks in the existing grid and potential infrastructure improvements which could alleviate those bottlenecks. Identify multiple alternative paths to achieve these goals in case the optimal construction project is blocked or infeasible.
- Virtual Power Plants: leverage AI to coordinate thousands of small-scale energy resources—like home batteries, rooftop solar, and EV chargers—to act as a single, large-scale power plant. When the grid is stressed, AI can instantly tap into these distributed batteries rather than turning on a larger but more remote power plant.
- Automated Fault Isolation: When electrical infrastructure fails (for example, a tree falls on a line or a transformer requires maintenance), AI-driven "self-healing" grids can re-route power around the damage. This limits the "dead time" where energy is lost and ensures the most efficient path for electricity is always maintained

Datasets & APIs:

- EIA Open Data API — US electricity generation, demand, and prices (free key, instant signup)
- CAISO OASIS — California ISO real-time prices, load, and renewables (no auth)
- GridStatus.io / `gridstatus` Python library — unified Python client for CAISO, PJM, ERCOT, MISO, ISO-NE, SPP
- NREL Developer APIs — solar irradiance (NSRDB), PVWatts, wind toolkit, utility rates
- PyPSA — open-source power system simulator; PyPSA-Eur has a full continental grid model
- LBNL US Data Center Energy Report — baseline data on data center load trends
- Pecan Street Dataport — residential smart-meter + rooftop solar data (free academic tier)
- PJM Data Miner 2 — clean REST endpoints for PJM load, LMPs, and ancillary services

Wargaming

Wargaming

Real wargames take months to design and days to run. Build tools or platforms to collapse that to hours, with AI running the scenario, the adversary, and the referee.

The idea: Wargaming has served as the ultimate "laboratory of the future" for the U.S. military, providing a low-cost, high-consequence environment to stress-test doctrine, refine multi-domain synchronization, and train leaders in the art of decision-making under uncertainty. However, the current wargaming enterprise is bottlenecked by analog inefficiencies: it often takes months of data collection and game design to build a high-fidelity scenario, and the "adjudication" phase—where white-cell experts determine the outcome of a move—can cause agonizing delays that disrupt the psychological flow of the exercise. These barriers of time and labor make high-quality wargaming an expensive luxury rather than a continuous tactical utility. This hackathon challenges you to shatter these constraints. We are looking for ways to leverage AI in wargames. This could be a new wargame played against an AI opponent. This could be a modular wargaming platform that can generate and adapt to new scenarios with minimal redesign. This could be an AI-framework that can autonomously generate complex "Order of Battle" datasets. You could also use Large Language Models to provide rapid adjudication of combat effects or create dynamic "Red Cell" adversaries that adapt to player behavior in real-time. Your mission is to do something to transform wargaming from a static, episodic event into a rapid, on-demand capability that matches the speed of modern conflict.

Judges:

SF	Boston	DC
Marcus Rossi	Dr. Yuna Wong	Dr. Yuna Wong
	Joseph Schwarzmann	Ming Ling

Example project directions:

- Autonomous red team: an LLM-based adversary that adapts its strategy based on blue force moves, using doctrine from Joint Chiefs publications
- Crisis escalation simulator: pull GDELT/ACLED event data to seed realistic starting conditions, then simulate how decisions cascade
- After-action reviewer: log all game states and generate a structured debrief — what worked, what didn't, what the doctrine says

* Note: We are not necessarily asking you to build a full game. You are welcome to, but other general ways to improve on the existing process of wargaming are also perfectly valid.

Datasets & APIs:

- GDELT 2.0 — global events and news graph updated every 15 minutes; BigQuery or

raw CSV

- ACLED — geocoded armed conflict and protest events worldwide (free API key)
- UCDP Georeferenced Event Dataset — Uppsala University's conflict event API, well-structured
- Global Terrorism Database — 200K+ incidents from 1970–2020, good for scenario seeding
- OpenStreetMap + Overpass API — query any global map data (roads, terrain features, infrastructure)
- OpenTopography (SRTM DEMs) — 30m/90m terrain elevation data for realistic terrain modeling
- Joint Chiefs Doctrine Library — all unclassified joint publications (JP 1-0, 3-0, 5-0)
- CSIS Analysis Library — open-access strategy papers and wargaming reports

GenAI.mil

GenAI.mil

The US military runs on paperwork. Build the AI assistant that makes the rank-and-file faster, smarter, and less buried in bureaucracy, and does it offline.

The idea: While modern defense often focuses on the "digital battlefield," the most immediate friction for the three million men and women in uniform often occurs in the "administrative trenches." From navigating thousands of pages of convoluted regulations and policies to the grueling manual labor of filling out paperwork for housing, maintenance, or housing, the "bureaucratic tail" of the military significantly drains mission readiness. The launch of the GenAI.mil portal provides a secure environment to weaponize AI against this inefficiency. The challenge for this hackathon is to build an AI-enabled software application that solves the everyday problems of the rank-and-file and streamlines the logistical hurdles that keep warfighters behind desks instead of in the field.

A good starting point: Pick one user persona (e.g., a junior NCO planning a training trip) and build an end-to-end solution for them. Set up a RAG pipeline over a small corpus of Army publications or Field Manuals (FMs). Get accurate retrieval working before adding form generation or logistics planning.

Judges:

SF	Boston	DC
David Vernal	Dr. Ho-Chit Liu	Kevin McQueary
Neeraj Chandra	Dr. Sanjeev Mohindra	Raj Panth
		Stuart Wagner

Example project directions:

- Regulation navigator: ingest Army Regulations and Field Manuals into a vector store; answer "what does AR 600-8-10 say about leave accrual?" accurately
- Form auto-filler: take a natural language request ("I need to file a DA 31 for 10 days leave starting June 3") and populate the correct form fields
- TDY planner: pull Joint Travel Regulations and GSA rates to generate a compliant travel itinerary with per diem calculations
- Contract intel tool: query USAspending or SAM.gov to surface relevant past awards and help a contracting officer understand the landscape

Datasets & APIs:

- DTIC Public STINET — DoD technical reports and TTPs, full-text searchable
- Army Publishing Directorate — all ARs, ADPs, FMs, and DA forms (bulk downloadable PDFs)

- Air Force e-Publishing — AFIs, AFMANs, and AF forms
- Joint Travel Regulations — the canonical DoD travel and per diem rulebook
- GSA Open APIs — federal procurement catalog and per diem rates
- SAM.gov Public API — contract opportunities and entity registrations (no auth for public data)
- USAspending.gov API — all federal contracts and spending, fully open
- Federal Register API — regulations, executive orders, notices
- eCFR Bulk Data — CFR Title 32 (National Defense) and Title 48 (Federal Acquisition) as XML/JSON